Jiaxin (Priscilla) Chen

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EDUCATION

Calvin University | Grand Rapids, MI
B.S. in Data Science & B.A. in Psychology August 2023 - (Expected) May 2027

- **GPA:** 3.972/4.00 (with Honors)
- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Applied Data Analysis, Data Structures and Algorithms, Database Management Systems, Optimization, Human-Centered AI, Visualizing Data and Models, Predictive Analytics, Computational Bayesian Statistics, Probability and Statistics

SKILLS

Programming Languages & Data: Python, SQL (MySQL), R, C#, C++; pandas, NumPy, scikit-learn, statistical modeling, Bayes
Systems & Tools: Linux, AWS, Git, Flask, RStudio, VS Code, Tableau
AI & ML: PyTorch, Hugging Face, model fine-tuning, LoRA/PEFT, LLM prompt engineering, tool calling, skills, agentic workflows, MCP

RESEARCH EXPERIENCE

Conflict-Resolved LoRA Merging for Copyright-Aware LLM Updates May 2026 – Present
Advisor: Prof. Xiangyu Zhao (City University of Hong Kong)

- Benchmark LLM memorization and copyright compliance with Hugging Face models, evaluating verbatim reproduction using LCS, ROUGE-L, and refusal rates.
- Designed Selective Sign Merging for LoRA adapters, resolving conflicting weight updates by adaptively preserving dominant contributions.
- Extended the method with SVD masked gating, using task-specific activations and local errors to calibrate singular-component corrections.
- Outperformed baseline model merges, on all evaluation metrics, improving unlearning by up to **13%** and relearning by up to **4%**
- Preparing the work for submission to ACL 2027.

Throughput Modeling for Internet Speed Test Sept. 2025 – Apr. 2026
Advisor: Prof. Rocky Chang (Calvin University)

- Analyzed approximately 300,000 rows of internet speed-test data across 100 tests using Linux shell pipelines and Python-based analytics workflows, integrating statistical analysis and machine learning clustering techniques for anomaly detection.
- Evaluated the robustness of internet speed tests using Ookla's Speedtest® under varying network conditions to identify potential measurement bias and improve fairness of performance.
- Co-authored a paper as second author, currently under review at ACM SIGMETRICS.

Residual Autocorrelation Diagnostics for Mixed-Effects Models Jun. 2025 – Mar. 2026
Advisor: Prof. Stacy DeRuiter (Calvin University)

- Conducted a simulation study to examine unexpected negative autocorrelation values at small lags in mixed-effects regression models with few observations per group.
- Identified the statistical mechanism behind this artifact and proposed guidelines for interpreting ACF diagnostics in repeated-measures regression models.
- First-authored a paper accepted to ACMS Conference Proceedings 2026.

PROJECT EXPERIENCE

AMIA Kaggle Competition, Disease Detection on Chest X-rays Jan. 2026 – Mar. 2026

- Fine-tuned an RT-DETR real-time object detection model for X-ray disease detection, ranking 10th out of 218 teams.
- Developed an end-to-end pipeline using NumPy and PyTorch for data preprocessing, model training, and evaluation; integrated Automatic Mixed Precision (AMP) and Distributed Data Parallel (DDP).
- Optimized training strategies, including early stopping, learning rate scheduling, and inference threshold tuning, to improve model performance and stability.
- Conducted systematic error analysis on model predictions to inform model iterations.

NVIDIA DGX Spark Hackathon: Multimodal AI Art Studio

- Co-developed an AI art studio that transforms children's drawings into animations, interactive 3D models, and narrated storybooks; earned a top-10 finish.
- Integrates 13 modular AI skills with multimodal models on NVIDIA DGX Spark, incorporating content screening and output-quality checks.
- Evaluated feedback and optimized model inference, reducing checked conversational response latency from 40–80 seconds to 2–8 seconds.

AI Development: Course Advisor Bot Aug. 2025 – Dec. 2025

- Developed a Python chatbot for personalized course recommendations, integrating OpenAI GPT APIs, supporting multi-turn dialogue, context tracking, and multilingual interaction with approximately 1s average latency per prompt.
- Led user testing with 22 participants that revealed LLM hallucinations recommending nonexistent courses.
- Implemented LLM tool calling with MCP and scraped the course catalog into a queryable database to address LLM hallucinations.

Computational Bayesian Inference of Currency Exchange Rate Jan. 2025 – May 2025

- Built a Bayesian model to infer the USD/CNY exchange rate using macroeconomic predictors.
- Preprocessed financial time-series data with missing value imputation, first-difference calculation, and standardization.
- Performed posterior inference using MCMC sampling and conducted convergence diagnostics to evaluate model fit and parameter uncertainty.

PUBLICATION & PRESENTATION

Kosters B., **Chen, J.**, Chang R. "FATHOM: A Framework for Adaptive Throughput Modeling in Browser-Based Speed Tests." Under Review at *ACM Sigmetrics,* 2026.

Chen, J., Crisman, M., & DeRuiter, S. L. "Using the Residual Autocorrelation Function to Verify Independence (and When It Doesn't Work)." Accepted to *ACMS Conference Proceedings 2026*.
Chen, J., Crisman, M., & DeRuiter, S. L. "Data Consulting Hub." *Electronic Undergraduate Statistics Research Conference*. Oral Presentation, Virtual. November 7, 2025.
Chen, J., Crisman, M., & DeRuiter, S. L. "Data Consulting Hub." *Calvin University STEM Summer Research Poster Fair*. Poster Presentation, Grand Rapids, MI. October 31, 2025.

LEADERSHIP EXPERIENCE

Tech Lead, Google Developer Groups on Campus (GDG on Campus) Aug. 2026 – Present
- Plan technical workshops and hands-on learning activities using Google developer resources for students interested in software development and AI.

CS Lab Teaching Assistant & Grader, Calvin University Dec. 2024 – Present
Supervisor: Prof. Rocky Chang, Prof. Ken Arnold, Prof. Henry Plantinga
- Assisted in computer science labs for CS108 and CS106: Introduction to Computing (Python), helping students debug code and understand programming concepts.
- Graded assignments for CS 212: Data Structures and Algorithms (C#) and provided feedback for further improvements.

Data Consulting Hub Research Assistant, Calvin University Jun. 2025 – Aug. 2025
Supervisor: Prof. Stacy DeRuiter
- Served as a data consultant for 13 research groups across STEM and social sciences, including HIV laboratory experiments and informal-caregiver surveys.
- Led end-to-end data analysis workflows, including data cleaning, statistical modeling, data visualization, and reporting.
- Contributed to conference presentations and research posters, demonstrating applications of statistical analysis.

AWARDS

McQueen Data Science Scholarship (top 1%) Jun. 2024, Jun. 2025, Jun. 2026
Dean's List May 2024, Dec. 2024, May 2025, Dec. 2025, May 2026
Trustee's Scholarship (Highest School-Wide Merit-Based Scholarship) Jun. 2023
Mosaic Award & International Assistant Grant Jun. 2023